Multi-Modal Fault Diagnosis on the CWRU Ten-Class Task: Expanded Methods on Time–Frequency Encodings and CNN Backbones

Zixu Wang

***Abstract*—We study a ten-class bearing fault diagnosis task on the Case Western Reserve University (CWRU) dataset. Beyond reporting accuracies for five encodings (CWT, STFT, GADF, MTF, RP) and three backbones (CNN, ResNet-18, RepLKNet), this expanded manuscript details the *principles, equations, and practical hyperparameters* of each method, and provides a multi-modal fusion recipe (image + sequence + cross-attention). On our splits, ResNet-18 attains 100% with CWT and STFT; RepLKNet reaches 100% with STFT and $99.11\,\%$ with CWT.**

***Index Terms*—Bearing fault diagnosis, CWRU, STFT, CWT, GADF, MTF, recurrence plot, ResNet-18, RepLKNet, multimodal fusion.**

I. Introduction

Vibration signals carry transient bursts, harmonics, and modulation sidebands when defects occur. Converting 1-D signals into 2-D *time–frequency* textures allows CNNs to learn spatial patterns that mirror what human analysts read from spectrograms or scalograms.

On the widely used CWRU benchmark [1], we compare popular encodings and backbones, and present a multimodal design (image + raw sequence) that remains compatible with domain adaptation. Our code uses standard preprocessing and training for reproducibility.

Contributions. (i) a controlled comparison of 5 encodings $\times$ 3 backbones; (ii) a simple, reproducible training recipe; and (iii) a fusion design that is compatible with domain adaptation. Specifically, we adopt ResNet-18 and a recent large-kernel CNN as backbones; draw on wavelet theory and discrete-time signal processing for time–frequency analysis; include GADF/MTF and recurrence plots as alternative 2-D encodings; consider multimodal fusion; and outline adversarial and kernel-based domain adaptation extensions.

II. Background and Methods

We denote a windowed signal $x = \{x_t\}_{t=1}^{T}$ (typically $T{=}1024$) sampled at rate f_s. Each encoding maps x to an image $I \in \mathbb{R}^{H\times W}$ later fed to a CNN f_θ; training minimizes cross-entropy.

Coursework manuscript.

A. Short-Time Fourier Transform (STFT)

With an analysis window $w(\cdot)$ of length L and hop H,

$$S(t,\omega) = \int x(\tau)\, w(\tau - t)\, e^{-j\omega\tau}\, d\tau, \quad \mathrm{Spec} = |S|. \tag{1}$$

Larger L improves frequency resolution but blurs impulses; smaller L sharpens transients but coarsens frequencies [2]. In practice we use a Hann window, $L/T \in [0.15, 0.30]$, hop $H = L/4{\sim}L/8$, magnitude scaling and $\log(1 + \alpha|S|)$ compression.

B. Continuous Wavelet Transform (CWT)

For mother wavelet ψ, scale a, and shift b,

$$W(a,b) = \int x(t)\, \psi^*\!\left(\frac{t-b}{a}\right) dt, \quad \mathrm{Scalogram} = |W|. \tag{2}$$

The pseudo-frequency is $f(a) = \frac{f_c}{a\,\Delta t}$ where f_c is the wavelet center frequency and $\Delta t = 1/f_s$ [3]. We adopt Morlet/complex-Morlet with 64–128 logarithmic scales and per-image min–max normalization.

C. Gramian Angular Difference Field (GADF)

GADF encodes global pairwise angular differences after mapping the series onto the unit circle; together with Markov Transition Field (MTF) it provides image-like representations of time series [4]. Steps: (1) normalize x to $\tilde{x} \in [-1, 1]$ and set $\phi_t = \arccos(\tilde{x}_t)$; (2) $G_{ij} = \cos(\phi_i - \phi_j)$; (3) optionally reindex by time to keep diagonals near the main temporal flow and resize to $H{\times}W$. GADF emphasizes *global correlation geometry* and pairs well with residual networks but is sensitive to scaling and resizing.

D. Markov Transition Field (MTF)

MTF captures *state-transition statistics* at every temporal pair. Steps: (1) quantize $\tilde{x}$ into Q bins and estimate the first-order Markov matrix P with $P_{uv} = \Pr(s_{t+1}{=}v \mid s_t{=}u)$; (2) build M with $M_{ij} = P_{s_i s_j}$. Larger Q preserves detail but increases noise; typical $Q \in [8, 32]$. MTF may smooth sharp impulses and underperform on strongly impulsive faults.

E. Recurrence Plot (RP)

RP visualizes when the trajectory returns near previously visited states [5]. With phase-space embedding $\mathbf{z}_t = [x_t, x_{t-\tau}, \ldots, x_{t-(m-1)\tau}]$, thresholded RP uses $R_{ij} = \mathbf{1}\{\|\mathbf{z}_i - \mathbf{z}_j\| < \varepsilon\}$; unthresholded variants use distances. Diagonal lines indicate periodicity; vertical/horizontal blocks indicate laminar states. Proper (m, τ, ε) is key; we adopt m=2$\sim$3 and τ near the first autocorrelation minimum.

F. Backbone CNNs

Shallow CNN (baseline): Four Conv–Norm–ReLU blocks with 3×3 kernels and stride/pooling for 4$\sim$8× downsampling, followed by *global average pooling (GAP)* and a linear head. Lightweight but limited receptive field and capacity.

ResNet-18: Residual blocks compute

$$\mathbf{y} = \mathcal{F}(\mathbf{x}; \theta) + \mathbf{x}, \qquad \mathcal{F} = \text{Conv–Norm–ReLU–Conv–Norm}. \tag{3}$$

Residual connections stabilize gradients and enable deeper receptive fields. We use ResNet-18 as a strong backbone [6].

RepLKNet (large-kernel CNN): Large depth-wise kernels (e.g., k=31$\sim$61) enlarge the *effective* receptive field:

$$\mathbf{y} = \mathrm{BN}\left(\mathrm{DWConv}_k(\mathbf{x}) + \sum_i \mathrm{Conv}^{(i)}_{1\times1}(\mathbf{x})\right). \tag{4}$$

Branches can be re-parameterized at inference to a single equivalent convolution. Large kernels capture long-range structures in STFT/CWT but may over-smooth encodings that already embed global geometry (e.g., GADF) unless normalization is tuned [7].

G. Training Objective and Regularizers

We minimize cross-entropy with optional label smoothing. For cross-condition robustness, we may add:

$$\text{DANN: } \min_{\theta_f, \theta_c} \max_{\theta_d} \; \mathcal{L}_{\text{cls}} - \lambda\, \mathbb{E}[\log D(f(x))], \tag{5}$$

$$\text{MMD: } \mathrm{MMD}^2 = \left\| \tfrac{1}{n}\sum_i \phi(f(x_i^s)) - \tfrac{1}{m}\sum_j \phi(f(x_j^t)) \right\|^2_{\mathcal{H}}. \tag{6}$$

These follow domain-adversarial learning and kernel two-sample testing, respectively [8], [9].

III. Multimodal Fusion: Image + Sequence

While the tabled results are single-modality, our deployable model fuses a 2-D image branch with a 1-D sequence branch [10].

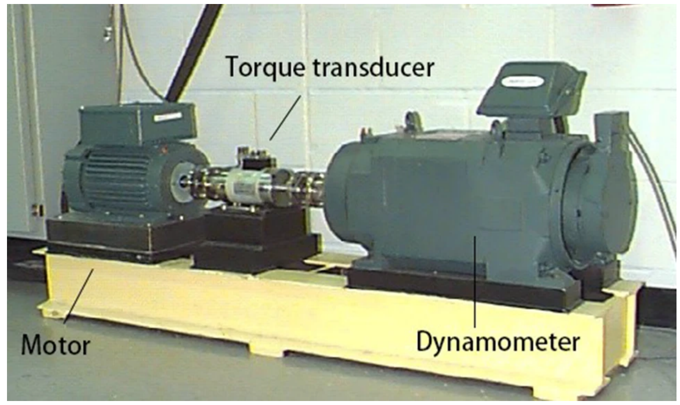

Fig. 1. Bearing test platform.

A. Sequence Branch (Conv1d + GN + BiGRU)

We first apply Conv1d+GroupNorm to denoise and expand channels, then a bidirectional GRU. For the GRU with input $\mathbf{u}_t$:

$$\mathbf{r}_t = \sigma(W_r\mathbf{u}_t + U_r\mathbf{h}_{t-1}), \quad \mathbf{z}_t = \sigma(W_z\mathbf{u}_t + U_z\mathbf{h}_{t-1}), \tag{7}$$

$$\tilde{\mathbf{h}}_t = \tanh(W_h\mathbf{u}_t + U_h(\mathbf{r}_t \odot \mathbf{h}_{t-1})), \tag{8}$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t. \tag{9}$$

The BiGRU concatenates forward/backward states and applies temporal pooling to yield $z_{\text{seq}} \in \mathbb{R}^d$.

B. Cross-Attention Fusion

Let $Z_{\text{img}} \in \mathbb{R}^{N\times d}$ and $Z_{\text{seq}} \in \mathbb{R}^{M\times d}$ be tokenized features. We compute multi-head attention from image→sequence and sequence→image, then concatenate:

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}(QK^\top/\sqrt{d})V, \tag{10}$$

$$\tilde{Z}_{\text{img}} = \mathrm{Attn}(Z_{\text{img}}W_Q, Z_{\text{seq}}W_K, Z_{\text{seq}}W_V), \tag{11}$$

$$\tilde{Z}_{\text{seq}} = \mathrm{Attn}(Z_{\text{seq}}W'_Q, Z_{\text{img}}W'_K, Z_{\text{img}}W'_V), \tag{12}$$

$$z_{\text{fuse}} = \mathrm{FFN}\Big([\mathrm{GAP}(\tilde{Z}_{\text{img}});\ \mathrm{GAP}(\tilde{Z}_{\text{seq}})]\Big). \tag{13}$$

Finally, $\hat{y} = \mathrm{softmax}(Wz_{\text{fuse}} + b)$.

IV. Datasets and Experimental Setup

Fig. 1 shows the experimental platform employed to generate rolling bearing fault data. Deep groove ball bearings (6205-2RS JEM SKF) were placed on the shaft of a 1.5 kW motor. The middle module was a torque sensor to collect motor speed and horsepower data. The final module was a dynamometer to measure power. There were also various control electronics not specifically shown in the figure.

Training and test data were derived from accelerometers placed at the drive end (DE) sampled at 48 kHz. Motor power and rotation were fixed at 3 hp and 1730 rpm, respectively. Three bearing failure types were considered: rolling, inner race, and outer race faults. Each fault type was induced in the rolling bearings by electro-discharge with different fault diameters (0.007, 0.014, and 0.021 inch). Thus, there were 10 faults in total: 3 fault types × 3 defect sizes + 1 normal signal; each with different and characteristic vibration signals.

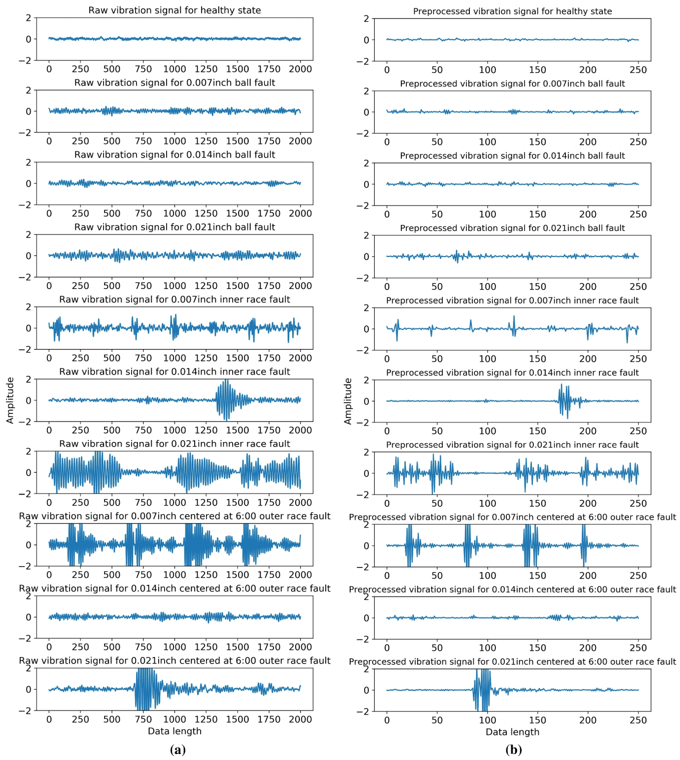

Fig. 2. Representative vibration signals.

Fig. 2 shows typical raw vibration signals for the 9 fault states (ball fault, inner race fault, and outer race fault, each with three defect sizes: 0.007, 0.014, and 0.021 inch) and the normal state. We selected partial vibration acceleration signals of 10 bearing states and plotted them. To ensure effectiveness and robustness of the proposed model, each bearing state (including healthy) included 240 samples acquired from accelerometers in the time domain, split into 70% for training and 30% for testing. Windows of $T = 1024$; per-window z-score.

Encodings. STFT/CWT/GADF/MTF/RP are rendered to 224×224 grayscale with range $[0, 1]$. Key parameters are summarized in Table I.

Backbones. Shallow CNN, ResNet-18, and RepLKNet.

Training. AdamW, cosine decay, batch 64, early stopping; light geometric/intensity augmentation for images. Default hyperparameters are listed in Table II.

V. Results

Backbone averages. CNN 82.69, ResNet-18 96.34, RepLKNet 90.63.

Encoding averages. CWT 96.73, STFT 94.94, RP 90.05, GADF 87.35, MTF 80.36.

VI. Discussion

Encodings. STFT/CWT best capture localized energy of impulsive faults; MTF is most sensitive to quantization Q and may over-smooth; GADF benefits from residual features and careful normalization; RP requires good (m, τ, ε) to avoid either sparsity or saturation.

TABLE I
Time–frequency/image encoding parameters (defaults used in our runs).

Method	Key parameters	Default setting
STFT	Window L, hop H, n_{fft}, compression $\log(1+\alpha\|S\|)$; resize to 224×224; per-image min–max to $[0, 1]$.	L=256 ($\approx 0.25T$), H=64, n_{fft}=512, α=10
CWT	Wavelet (complex Morlet), #scales, log-spaced scales, pseudo-frequency map; resize to 224×224; per-image min–max.	cmor1.5–1.0; 128 scales; log-spaced
GADF	Normalize x to $[-1, 1]$ (min–max); angles $\phi_t = \arccos(\tilde{x}_t)$; $G_{ij} = \cos(\phi_i - \phi_j)$; resize to 224×224.	$[-1, 1]$ min–max; bilinear resize
MTF	Quantization bins Q; Markov matrix P; $M_{ij} = P_{s_i s_j}$; optional Gaussian smoothing; resize to 224×224.	Q=16; light smoothing (optional)
RP	Embedding (m, τ); threshold ε or percentile; recurrence rate control; resize to 224×224.	m=2∼3, τ at first ACF minimum; ε at 5% percentile

TABLE II
Training hyperparameters and preprocessing defaults.

Component	Setting
Sampling / window	f_s=12.00 kHz; T=1024 samples/window
Batch size	64
Optimizer	AdamW
Initial LR	Shallow CNN: 3×10^{-4}; ResNet-18: 1×10^{-4}; RepLKNet: 1×10^{-4}
Weight decay	1×10^{-4}
LR schedule	Cosine annealing; $\eta_{\min}$=1×10^{-6}; warmup 5 epochs
Epochs / early stopping	Max 100 epochs; patience 15 (val acc)
1-D preprocessing	Per-window z-score (channel-wise)
Image preprocessing	Per-image min–max to $[0, 1]$; size 224×224
Augmentations (imgs)	Random resize/crop ($\pm$10%), mild brightness/contrast jitter
Fusion (if used)	Heads h=4∼8, feature dim d=128∼256; DA losses on z_{fuse}

Backbones. ResNet-18 is a reliable default; RepLKNet excels on STFT/CWT but needs tuned norms on geometry-heavy encodings (e.g., GADF).

Multimodality. Under domain shift (speed/load), adding the sequence branch plus cross-attention typically improves robustness; DA losses can be attached to the fused space.

VII. Conclusion

CWT/STFT paired with ResNet-18 provide a simple, accurate baseline for CWRU ten-class classification; RepLKNet is competitive on STFT/CWT. We provide an extensible multimodal design for future cross-condition experiments.

TABLE III
ACCURACY (%) ON THE CWRU TEN-CLASS TASK.

Backbone	CWT	GADF	MTF	RP	STFT
CNN	91.07	83.48	70.54	83.53	84.82
ResNet-18	**100.00**	**98.66**	**87.50**	**95.54**	**100.00**
RepLKNet	99.11	79.91	83.04	91.07	**100.00**

REFERENCES

[1] W. A. Smith and R. B. Randall, "Rolling element bearing diagnostics using the case western reserve university data: A benchmark study," *Mechanical systems and signal processing*, vol. 64, pp. 100–131, 2015.

[2] A. V. Oppenheim, *Discrete-time signal processing*. Pearson Education India, 1999.

[3] I. Daubechies, *Ten lectures on wavelets*. SIAM, 1992.

[4] Z. Wang and T. Oates, "Imaging time-series to improve classification and imputation," *arXiv preprint arXiv:1506.00327*, 2015.

[5] J.-P. Eckmann, S. O. Kamphorst, and D. Ruelle, "Recurrence plots of dynamical systems," in *Turbulence, strange attractors and chaos*. World Scientific, 1995, pp. 441–445.

[6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[7] X. Ding, X. Zhang, J. Han, and G. Ding, "Scaling up your kernels to 31x31: Revisiting large kernel design in cnns," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11 963–11 975.

[8] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. March, and V. Lempitsky, "Domain-adversarial training of neural networks," *Journal of machine learning research*, vol. 17, no. 59, pp. 1–35, 2016.

[9] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, "A kernel two-sample test," *The journal of machine learning research*, vol. 13, no. 1, pp. 723–773, 2012.

[10] S. R. Stahlschmidt, B. Ulfenborg, and J. Synnergren, "Multimodal deep learning for biomedical data fusion: a review," *Briefings in bioinformatics*, vol. 23, no. 2, p. bbab569, 2022.